THENI KAMMAVAR SANGAM
COLLEGE OF ARTS AND SCIENCE (Co-Ed)
Approved by Govt. of Tamil Nadu | Affiliated to Madurai Kamaraj University
Recognized under section 2(f) of the UGC Act, 1956 | ISO 9001:2015 Certified institution
KODUVILARPATTI, THENI- 625 534.
04546 - 250882, 8925192579, www.tkscas.com, tkscgas@gmail.com

HTML QUICK STYLER

AI-Powered HTML Page Generator With IBM Granite 3.3 2B

Submitted by

1.Gobika V. (C3S29456)

2.Jeevana R. (C3S29457)

3.Kaleeswari M. (C3S2958)

4.Karpagam G. (C3S29460)

5. Kaviprabha V. (C3S29461)

Department of Computer Science

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1. Abstract

HTML Quick Styler is an intelligent system designed to automatically generate complete HTML web pages from user descriptions. Creating professional web pages manually requires knowledge of HTML, CSS, and design principles, which can be difficult for beginners.

This project simplifies the process by allowing users to describe the website they want in natural language. The system analyzes the input and automatically generates structured HTML code with modern CSS styling.

The system uses the IBM Granite 3.3 2B Instruct model to understand user requirements and produce responsive and well-designed web pages. A user-friendly interface built with Gradio allows users to generate and preview web pages easily.

2. Introduction

Web development requires writing structured HTML and CSS code to design attractive and responsive websites. However, beginners often find it difficult to create professional layouts and styles.

The HTML Quick Styler project aims to simplify web page creation using artificial intelligence. Users can simply describe their website requirements, and the system generates the corresponding HTML and CSS code automatically.

The system analyzes the user input, detects required sections such as hero, about, portfolio, or contact sections, and builds a responsive website layout.

3. Objectives of the Project

- To automatically generate HTML web pages from user descriptions
- To reduce the complexity of manual web development
- To provide responsive and professional CSS styling
- To generate modern web layouts with multiple sections
- To provide an easy-to-use interface for page generation
- To help beginners quickly build websites without deep coding knowledge

4. Technologies Used

Python – Main programming language.

IBM Granite 3.3 2B Instruct Model – Used to generate HTML structures.

Gradio – Used to create the web interface.

HTML5 and CSS3 – Used for webpage structure and styling.

JavaScript – Provides interactivity and animation.

Google Colab – Cloud environment used to run the project.

5. System Architecture

The system architecture consists of several modules working together to generate HTML pages.

User Interface Module
Users enter the page title and description.

Input Processing Module
Detects webpage sections.

HTML Generation Module
Creates HTML structure.

CSS Styling Module
Applies modern styles and responsive design.

Preview Module
Shows the generated webpage.

6.Database Schema

1. Users Table

This table stores the details of users who use the system.

Fields:

- id (Primary Key)
- name
- email
- password
- signup_date
- status

2. Projects Table

This table stores the details of website projects created by users.

Fields:

- id (Primary Key)
- user_id (Foreign Key)
- project_title
- description
- created_date
- last_updated

3. Webpages Table

This table stores the generated HTML page information.

Fields:

- id (Primary Key)
- project_id (Foreign Key)
- page_name
- html_code
- css_code
- created_date

4. Templates Table

This table stores the website templates used by the system.

Fields:

- id (Primary Key)

- template_name
- template_type
- html_structure
- css_style
- created_date

5. Sections Table

This table stores the different sections of a webpage such as Hero, About, Contact, etc.

Fields:

- id (Primary Key)
- page_id (Foreign Key)
- section_name
- section_content
- section_order

7. Implementation Methodology

The project is implemented in several stages.

Environment Setup

Install required libraries.

Model Configuration

Load IBM Granite model.

HTML Generation Logic

Detect sections and generate templates.

CSS Styling Engine

Apply responsive design and layouts.

Interface Development

Build Gradio interface.

8. System Features

- Automatic HTML page generation
- Responsive CSS styling
- Intelligent section detection
- Multiple color palette options
- Live preview of generated pages
- User-friendly interface

9.Testing and Results

Input:
Create a portfolio website with hero section, about section, project gallery, and contact form.

Output:
The system generates a complete HTML page including hero section, about section, portfolio gallery, and contact form with responsive design.

10. Future Scope

- Custom color picker
- Integration with React
- Automatic JavaScript generation
- Template library expansion
- Export generated HTML files

11. Conclusion

HTML Quick Styler simplifies website creation using artificial intelligence. It allows users to generate professional web pages quickly without deep coding knowledge.